Jaisingh Pal

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Google Scholar · LinkedIn

Ph.D. Research Scholar · NanoDC Lab · Electrical Engineering, IIT Gandhinagar

RESEARCH INTERESTS

Device physics & TCAD simulation · Wafer-level characterization · Comprehensive analysis of Complementary FET (CFET) and Nanosheet architectures

EDUCATION

Ph.D., Electrical Engineering IIT Gandhinagar — Jul 2022 Present
Research Scholar, NanoDC Lab · Gandhinagar, India

B.Tech., Electronics & Telecommunication VIIT, Pune — 2019 2022
CGPA 9.17 / 10

Diploma, Electronics & Telecommunication Sandip Foundation — 2017 2019
85.47% · Maharashtra, India

Class X Vidya Prabodhini Prashala — 2015 2016
75.2% · Nashik, India

EXPERIENCE

Lab Administrator, Wafer Characterization Lab IIT Gandhinagar — Jul 2022 Present
Managed wafer tools, performed data analysis, and conducted material characterization to support semiconductor research and development.

Teaching Assistant IIT Gandhinagar

- Microfabrication & Semiconductor Processes (ES626) · Microelectronics Lab (EE618)
- Principles & Applications of Electrical Engineering (ES116) · Writing (FP602)

Summer Intern Sarvaa Research & Innovation Pvt. Ltd., Bengaluru — Apr Nov 2021
Designed power systems for wearable devices. Guide: Shwetha R. (Director).

Summer Intern E-Smart Pvt. Ltd., Nashik — Apr Jun 2018
Electronics testing, troubleshooting, and performance analysis of semiconductor components. Guide: Dipak Deshmukh (Senior Engineer).

RESEARCH & DEVELOPMENT WORK

Parasitic Capacitance in CFETs
Developed an analytical model for parasitic capacitance in CFETs, validated through TCAD simulations, identifying separator capacitance as a key contributor.

3D Simulation of CFET
Conducted 3D TCAD simulations to study power efficiency, leakage, and 3D-integration potential.

2D Simulation of HBT
Simulated a 2D HBT in TCAD, analysing emitterbase and basecollector heterojunctions for enhanced performance.

Radiation-Hardened 14T SRAM

Designed a radiation-hardened 14T SRAM bit-cell, optimising speed, power, and reliability for space applications.

Analog Circuit Design

Developed a low-power 2-stage op-amp and band-gap reference in SCL 180 nm technology; analysed multiple amplifier configurations.

PUBLICATIONS

1. A. Singh, **Jaisingh Pal**, Om Maheshwari, and Nihar R. Mohapatra. "Comprehensive Study of Parasitic Capacitance in CFETs: An Analytical Perspective." *IEEE Transactions on Electron Devices*, 2025. DOI: 10.1109/TED.2025.3585907.
2. Prateek Sharma, **Jaisingh Pal**, and Sandip Lashkare. "Bit-wise Logical Operations using Capacitor-less Silicon-on-Insulator MOSFET for In-memory Computing." *2024 8th IEEE Electron Devices Technology & Manufacturing Conference (EDTM)*, IEEE, 2024.
3. **Jaisingh Pal** and Nihar R. Mohapatra. "Parasitic Capacitance Analysis in Tapered CFET Devices for Advanced Technology Nodes." *IEEE International Conference on Emerging Electronics (ICEE)*, 2025.

TECHNICAL SKILLS

EDA Tools: Synopsys TCAD, Cadence Virtuoso, Xilinx ISE Design Suite, MATLAB, Velox, EasyExpert
Programming: Python, Verilog, C
Practical: Wafer handling, wafer-tool operation, device characterization, data analysis, lab management

RELEVANT COURSEWORK

1. CMOS Analog IC Design
2. CMOS Analog IC Design Lab
3. Microfabrication
4. VLSI Design
5. Physics of Transistors
6. Heterostructures & Devices
7. Memory Device Technologies & Applications
8. Machine Learning
9. Writing